



CCODEL
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GLOBAL LEADERS

Computer Programming I (Java 1)

COS221

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MODULE ONE: INTRODUCTION TO COMPUTER PROGRAMMING

UNIT ONE: OVERVIEW OF COMPUTER PROGRAMMING

OVERVIEW OF COMPUTER PROGRAMMING

Learning Objectives

By the end of this unit, you will have been introduced to:

- **the concept of computer programming and its significance.**
- **various programming paradigms and the programming languages associated with each.**
- **basic concepts such as attributes, methods, classes, and objects using examples from Java programs.**

Learning Outcomes

At the end of this unit, you should be able to:

- **explain computer programming and its significance.**
- **identify various programming paradigms and their associated programming languages.**
- **identify basic concepts like attributes, methods, class and objects in a written Java program.**

Introduction

- **Computer programming is the process of writing instructions that tell a computer how to perform tasks. It plays a vital role in software development and drives innovation in areas like artificial intelligence, web development, data analysis, and automation.**
- **This unit introduces the core principles of programming, explores different paradigms and their applications, and guides you through writing your first Java program.**

What is Computer Programming?

- **Programming is the process of giving a computer instructions to perform tasks using code. A programmer writes these instructions in specialized languages like Java, Python, or C++.**
- **Just as a natural language learner learns by reading books and materials written in that language, a programmer studies existing code to understand its structure and logic. This foundation helps in writing effective software.**

Why is Programming Important?

- **Task Automation:** Programming allows repetitive tasks to be automated, saving time and reducing human errors.
- **Software Development:** It enables the creation of websites, apps, and enterprise solutions that drive businesses.
- **Problem-Solving:** Programming teaches logical thinking and problem-solving, which are crucial skills in various fields.
- **Career Growth:** Programming offers diverse high-paying opportunities in software engineering, data science, etc.

Programming Paradigms

A programming paradigm refers to a fundamental style or approach to writing computer programs. The following are some programming paradigms:

- **Imperative Programming**
- **Functional Programming**
- **Declarative Programming**
- **Logic-Based Programming**
- **Object-Oriented Programming (OOP)**

Imperative Programming

- This is a step-by-step approach where the program tells the computer how to achieve a task.
- Key concepts here include the use of variables, loops and conditionals.

This C program calculates and prints the sum of two integers.

```
int main () {  
    int x = 5;  
    int y = 10;  
    int sum = x + y;  
    Printf("Sum : %d", sum);  
    return 0;  
}
```

Functional Programming

Functional programming focuses on what should be computed rather than how. Functions are treated as first-class citizens. Key concepts include using pure functions, avoid changing variables (immutability).

- **Examples includes Haskell, Lisp, JavaScript (functional style), etc.:**

```
function add(x, y) {  
    return x + y;  
}  
console.log(add(5, 10))
```

Declarative Programming

- **Declarative programming is a programming paradigm that focuses on what the program should accomplish rather than how to accomplish it. Unlike imperative programming, which requires step-by-step instructions, declarative code expresses the logic of computation without explicitly describing its control flow. Samples of Declarative Programming in SQL:**

```
SELECT name, age FROM students WHERE age > 18;
```

```
SELECT name FROM employees WHERE department = 'Sales';
```

Logic-Based Programming

- This is a style of declarative programming where computation is expressed in terms of formal logic. Instead of writing instructions, you write facts and rules, and the language engine infers answers using logical deduction. Samples of codes in in Prolog:

```
male (john).  
female (susan).  
likes (susan, pasta).  
parent (susan, alice).  
parent (john, bob).  
?- grandparent(john, susan).
```

Object-Oriented Programming (OOP)

This organizes codes using objects (Figure 1.2), which are instances of classes (Figure 1.1). These objects represent real-world entities and contain both:

- **Data (attributes or state)**
- **Methods (functions or behavior)**

The main idea of OOP is to model software as a collection of interacting objects, each responsible for its own data and logic.

Examples include Java, C++, Python (OOP style), etc. Code sample of OOP paradigm in Java:

A Quick Dive into the Concept of OOP

```
// Defining a class
class Car {
    String brand = "Toyota";    // Attribute
    int speed = 120;            // Attribute

    // Method to display car details
    void displayInfo() {
        System.out.println("Car brand: " + brand);
        System.out.println("Car speed: " + speed);
    }
}

public class Main {
    public static void main(String[] args) {
        Car carObject = new Car (); // Creating an object
        carObject.displayInfo();     // Calling method
    }
}
```

A Quick Dive into the Concept of OOP

```
class Dog {  
    // Attributes  
    String name;  
    int age;  
  
    // Method  
    void bark() {  
        System.out.println("Woof!");  
    }  
}
```

```
public class Main {  
    public static void main (String[] args) {  
        // Create a Dog Object  
        Dog myDog = new Dog();  
  
        // Assigning Value  
        myDog.name = "Buddy";  
        myDog.age = 3;  
  
        // Calling Method  
        myDog.bark();  
    }  
}
```

The code (left) defines a Dog class as a blueprint for creating dog objects with attributes like name and age. It also includes a bark() method, allowing all dog instances to share common behavior while having unique traits.

In-Text Questions (ITQs)

- 1. What is computer programming?**
- 2. Why is programming considered the backbone of software development?**
- 3. List three major benefits of learning programming.**

In-Text Answers (ITAs)

- 1. Computer programming is the process of writing instructions that enable a computer to perform specific tasks.**
- 2. Because it drives innovation in areas like artificial intelligence, web development, data analysis, and automation.**
- 3. a) Task automation to save time and reduce errors.**
b) Creation of software like websites and mobile apps.
c) Development of logical thinking and problem-solving skills.

Conclusion

- **Computer programming is a vital skill in today's digital world, enabling innovation, automation, and effective problem-solving.**
- **This unit has laid a strong foundation in programming concepts and paradigms, with a focus on understanding and applying basic Java principles.**
- **With this knowledge, learners are now equipped to write simple programs and explore more advanced programming techniques in future modules.**

Summary

- **This unit introduces the foundational concepts of computer programming, emphasizing its significance in modern technology and problem-solving.**
- **It explores various programming paradigms: imperative, functional, declarative, logic-based, and object-oriented, highlighting their unique approaches and use cases.**